

Type-3: Interval $(-\pi, \pi)$

1. Obtain Fourier series for $f(x) = e^x$ in $(-\pi, \pi)$. Hence derives the series for $\frac{\pi}{\sin h \pi}$

Solution: Let $f(x) = \frac{a_0}{2} + \sum a_n \cos nx + \sum b_n \sin nx$ (1)

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{\pi} \int_{-\pi}^{\pi} e^x dx = \frac{1}{\pi} [e^x]_{-\pi}^{\pi} = \frac{1}{\pi} [e^{\pi} - e^{-\pi}] = \frac{2}{\pi} \sinh \pi$$

$$\begin{aligned} a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx \\ &= \frac{1}{\pi} \int_{-\pi}^{\pi} e^x \cos nx dx = \frac{1}{\pi} \left[\frac{1}{1+n^2} \cdot e^x (\cos nx + n \sin nx) \right]_{-\pi}^{\pi} \\ &\quad \left[\because \int e^x \cos nx dx = \frac{1}{a^2+b^2} \cdot e^{ax} (a \cos bx + b \sin bx) \right] \\ &= \frac{1}{\pi} \cdot \frac{1}{1+n^2} [e^{\pi} \cos n\pi - e^{-\pi} \cos n\pi] \\ &= \frac{2 \cos n\pi}{\pi(1+n^2)} \cdot \frac{e^{\pi} - e^{-\pi}}{2} = \frac{2 \cos n\pi}{\pi(1+n^2)} \cdot \sinh \pi \\ &= \frac{(-1)^n}{\pi(1+n^2)} \cdot 2 \sinh \pi \end{aligned}$$

$$\begin{aligned} b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx \\ &= \frac{1}{\pi} \int_{-\pi}^{\pi} e^x \sin nx dx = \frac{1}{\pi} \left[\frac{1}{1+n^2} \cdot e^x (\sin nx - n \cos nx) \right]_{-\pi}^{\pi} \\ &\quad \left[\because \int e^x \sin nx dx = \frac{1}{a^2+b^2} \cdot e^{ax} (a \sin bx + b \cos bx) \right] \\ &= \frac{1}{\pi} \cdot \frac{1}{1+n^2} [-e^{\pi} n \cos n\pi + n e^{-\pi} \cos n\pi] \\ &= \frac{-2n \cos n\pi}{\pi(1+n^2)} \cdot \left[\frac{e^{\pi} - e^{-\pi}}{2} \right] = \frac{-2n \cos n\pi}{\pi(1+n^2)} \cdot \sinh \pi \\ &= -(-1)^n \frac{n}{\pi(1+n^2)} \cdot 2 \sinh \pi \end{aligned}$$

Putting these values in (1),

$$\begin{aligned} \therefore e^x &= \frac{2 \sinh \pi}{\pi} \left[\frac{1}{2} + \sum_{n=1}^{\infty} \frac{(-1)^n}{(1+n^2)} \cos nx - \sum_{n=1}^{\infty} \frac{n(-1)^n}{(1+n^2)} \sin nx \right] \\ &= \frac{2 \sinh \pi}{\pi} \left[\frac{1}{2} - \frac{1}{2} \cdot \cos x + \frac{1}{5} \cdot \cos 5x - \frac{1}{10} \cdot \cos 3x + \dots + \frac{1}{2} \cdot \sin x - \frac{2}{5} \cdot \sin 2x + \frac{3}{10} \cdot \sin 3x - \dots \right] \end{aligned}$$

$$\text{Put } x = 0, \frac{\pi}{\sinh \pi} = 2 \left\{ \frac{1}{5} - \frac{1}{10} + \frac{1}{17} - \dots \right\} \text{ or } \frac{\pi}{\sinh \pi} = 2 \left[\frac{1}{1+2^2} - \frac{1}{1+3^2} + \dots \right]$$

2. Find the Fourier series for $f(x)$, if $f(x) = \begin{cases} -\pi, & -\pi < x < 0 \\ x & 0 < x < \pi \end{cases}$

Hence deduce that $\sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots = \frac{\pi^2}{8}$

Solution: Let $f(x) = \frac{a_0}{2} + \sum a_n \cos nx + \sum b_n \sin nx$ (1)

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{\pi} \left[\int_{-\pi}^0 -\pi \cdot dx + \int_0^{\pi} x dx \right] = \frac{1}{\pi} \left[-\pi \{x\}_{-\pi}^0 + \left\{ \frac{x^2}{2} \right\}_0^{\pi} \right] = \frac{1}{\pi} \left[-\pi^2 + \frac{\pi^2}{2} \right] = -\frac{\pi}{2}$$

$$\begin{aligned} a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx \\ &= \frac{1}{\pi} \left[\int_{-\pi}^0 -\pi \cos nx dx + \int_0^{\pi} x \cos nx dx \right] = \frac{1}{\pi} \left[-\pi \left(\frac{\sin nx}{n} \right)_{-\pi}^0 + \left\{ x \left(\frac{\sin nx}{n} \right) - (1) \left\{ -\frac{\cos nx}{n^2} \right\} \right\}_0^{\pi} \right] \\ &= \frac{1}{\pi} \left[-\pi(0 - 0) + \left\{ \pi(0) + \frac{\cos n\pi}{n^2} - 0 - \frac{1}{n^2} \right\} \right] \\ &= \frac{1}{\pi} \cdot \left[\frac{(-1)^n}{n^2} - \frac{1}{n^2} \right] = \frac{1}{\pi n^2} [(-1)^n - 1] \end{aligned}$$

$$\begin{aligned}
b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx \, dx = \frac{1}{\pi} \left[\int_{-\pi}^0 -\pi \sin nx \, dx + \int_0^{\pi} x \sin nx \, dx \right] \\
&= \frac{1}{\pi} \left[-\pi \left(-\frac{\cos nx}{n} \right)_{-\pi}^0 + \left\{ x \left(-\frac{\cos nx}{n} \right) - (1) \left(-\frac{\sin nx}{n} \right) \right\}_0^{\pi} \right] \\
&= \frac{1}{\pi} \left[\pi \frac{(1 - \cos n\pi)}{n} + \left\{ \pi \left(-\frac{\cos n\pi}{n} \right) - 0 \right\} \right] \\
&= \frac{1}{\pi} \cdot \left[\frac{\pi}{n} (1 - \cos n\pi) - \frac{\pi}{n} \cos n\pi \right] = \frac{1}{n} [1 - 2 \cos n\pi] = \frac{1}{n} [1 - 2(-1)^n]
\end{aligned}$$

Putting these values in (1),

$$\begin{aligned}
f(x) &= -\frac{\pi}{4} + \frac{1}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^2} \cos nx + \sum_{n=1}^{\infty} \frac{1-2(-1)^n}{n} \sin nx \\
&= -\frac{\pi}{4} - \frac{2}{\pi} \left[\frac{\cos x}{1^2} + \frac{\cos 3x}{3^2} + \frac{\cos 5x}{5^2} + \dots \right] + \left[3 \cdot \frac{\sin x}{1} - \frac{1}{2} \sin 2x + \sin 3x - \dots \right] \quad \dots\dots\dots (2)
\end{aligned}$$

Now, $f(x)$ is discontinuous at $x = 0$.

$$\text{At point of discontinuity } x = c, \quad f(x) = \frac{1}{2} \left[\lim_{x \rightarrow c^-} f(x) + \lim_{x \rightarrow c^+} f(x) \right]$$

$$\therefore f(0) = \frac{1}{2} [-\pi + 0] = -\frac{\pi}{2}$$

Hence, putting $x = 0$ in (2),

$$\begin{aligned}
-\frac{\pi}{2} &= -\frac{\pi}{4} - \frac{2}{\pi} \left[\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots \right] \\
\therefore \frac{\pi^2}{8} &= \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots = \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2}
\end{aligned}$$

3. Find the Fourier Series for $f(x) = \begin{cases} \cos x, & -\pi < x < 0 \\ \sin x, & 0 < x < \pi \end{cases}$

Solution: Let $f(x) = \frac{a_0}{2} + \sum a_n \cos nx + \sum b_n \sin nx \quad \dots\dots\dots (1)$

$$\begin{aligned}
a_0 &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \, dx = \frac{1}{\pi} \left[\int_{-\pi}^0 \cos x \, dx + \int_0^{\pi} \sin x \, dx \right] \\
&= \frac{1}{\pi} \left[\{\sin x\}_{-\pi}^0 + \{-\cos x\}_0^{\pi} \right] = \frac{1}{\pi} [0 + (1 + 1)] = \frac{2}{\pi}
\end{aligned}$$

$$\begin{aligned}
a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx \, dx \\
&= \frac{1}{\pi} \left[\int_{-\pi}^0 \cos x \cos nx \, dx + \int_0^{\pi} \sin x \cos nx \, dx \right] \\
&= \frac{1}{2\pi} \left[\int_{-\pi}^0 2 \cos nx \cos x \, dx + \int_0^{\pi} 2 \cos nx \sin x \, dx \right] \\
&= \frac{1}{2\pi} \left[\int_{-\pi}^0 [\cos(n+1)x + \cos(n-1)x] \, dx + \int_0^{\pi} [\sin(n+1)x - \sin(n-1)x] \, dx \right] \\
&= \frac{1}{2\pi} \left[\left\{ \frac{\sin(n+1)x}{n+1} + \frac{\sin(n-1)x}{n-1} \right\}_{-\pi}^0 + \left\{ -\frac{\cos(n+1)x}{n+1} + \frac{\cos(n-1)x}{n-1} \right\}_0^{\pi} \right] \\
&= \frac{1}{2\pi} \left[\{0\} + \left\{ -\frac{\cos(n+1)\pi}{n+1} + \frac{\cos(n-1)\pi}{n-1} + \frac{1}{n+1} - \frac{1}{n-1} \right\} \right] \text{ if } n \neq 1 \\
\therefore a_n &= \begin{cases} 0 & \text{if } n \text{ is odd and } \neq 1 \\ -\frac{2}{\pi(n^2-1)} & \text{if } n \text{ is even} \end{cases}
\end{aligned}$$

When $n = 1$, we have

$$\begin{aligned}
a_1 &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos x \, dx \\
&= \frac{1}{\pi} \left[\int_{-\pi}^0 \cos x \cos x \, dx + \int_0^{\pi} \sin x \cos x \, dx \right]
\end{aligned}$$

$$\begin{aligned}
&= \frac{1}{\pi} \left[\int_{-\pi}^0 \left(\frac{1+\cos 2x}{2} \right) dx + \int_0^{\pi} \frac{\sin 2x}{2} dx \right] \\
&= \frac{1}{\pi} \left[\frac{1}{2} \left(x + \frac{\sin 2x}{2} \right)_{-\pi}^0 + \left(-\frac{\cos 2x}{4} \right)_0^{\pi} \right] \\
&= \frac{1}{\pi} \left[\frac{1}{2} \left\{ 0 - (-\pi - 0) - \left(\frac{1}{4} - \frac{1}{4} \right) \right\} \right] = \frac{1}{2}
\end{aligned}$$

$$\begin{aligned}
b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx \, dx \\
&= \frac{1}{\pi} \left[\int_{-\pi}^0 \cos x \sin nx \, dx + \int_0^{\pi} \sin x \sin nx \, dx \right] \\
&= \frac{1}{2\pi} \left[\int_{-\pi}^0 2 \cos x \sin nx \, dx + \int_0^{\pi} 2 \sin x \sin nx \, dx \right] \\
&= \frac{1}{2\pi} \left[\int_{-\pi}^0 [\sin(n+1)x + \sin(n-1)x] \, dx - \int_0^{\pi} [\cos(n+1)x - \cos(n-1)x] \, dx \right] \\
&= \frac{1}{2\pi} \left[\left\{ -\frac{\cos(n+1)x}{n+1} - \frac{\cos(n-1)x}{n-1} \right\}_{-\pi}^0 - \left\{ \frac{\sin(n+1)x}{n+1} - \frac{\sin(n-1)x}{n-1} \right\}_0^{\pi} \right] \\
&= \frac{1}{2\pi} \left[\left\{ -\frac{1}{n+1} - \frac{1}{n-1} + \frac{\cos(n+1)\pi}{n+1} + \frac{\cos(n-1)\pi}{n-1} \right\} - 0 \right] \text{ if } n \neq 1 \\
\therefore b_n &= \begin{cases} 0 & \text{if } n \text{ is odd and } \neq 1 \\ -\frac{2n}{\pi(n^2-1)} & \text{if } n \text{ is even} \end{cases}
\end{aligned}$$

When $n = 1$, we have

$$\begin{aligned}
b_1 &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin x \, dx \\
&= \frac{1}{\pi} \left[\int_{-\pi}^0 \cos x \sin x \, dx + \int_0^{\pi} \sin^2 x \, dx \right] \\
&= \frac{1}{\pi} \left[\int_{-\pi}^0 \frac{\sin 2x}{2} \, dx + \int_0^{\pi} \frac{1-\cos 2x}{2} \, dx \right] \\
&= \frac{1}{\pi} \left[\left\{ -\frac{\cos 2x}{2} \right\}_{-\pi}^0 + \frac{1}{2} \left\{ x - \frac{\sin 2x}{2} \right\}_0^{\pi} \right] \\
&= \frac{1}{\pi} \left[-\left(\frac{1}{2} - \frac{1}{2} \right) + \frac{1}{2}(\pi - 0) \right] = \frac{1}{2}
\end{aligned}$$

Putting these values in (1)

$$\begin{aligned}
f(x) &= \frac{a_0}{2} + a_1 \cos x + [a_2 \cos 2x + a_4 \cos 4x + \dots] + b_1 \sin x + [b_2 \sin 2x + b_4 \sin 4x + \dots] \\
&= \frac{1}{\pi} + \frac{1}{2} \cos x - \frac{2}{\pi} \left[\frac{\cos 2x}{3} + \frac{\cos 4x}{15} + \dots \right] + \frac{1}{2} \sin x - \frac{2}{\pi} \left[\frac{2 \sin 2x}{3} + \frac{4 \sin 4x}{15} + \dots \right] \\
&= \frac{1}{\pi} + \frac{1}{2} (\cos x + \sin x) + \frac{2}{\pi} \left[\sum_{n=1}^{\infty} \frac{1}{(1-4n^2)} \cos 2nx + \sum_{n=1}^{\infty} \frac{2n}{(1-4n^2)} \sin 2nx \right]
\end{aligned}$$

4. Find the Fourier expansion of $f(x) = x^2$, $-\pi \leq x \leq \pi$, and hence, prove that

$$\begin{aligned}
\text{(i)} \quad \frac{\pi^2}{6} &= \sum_{n=1}^{\infty} \frac{1}{n^2}, & \text{(ii)} \quad \frac{\pi^2}{12} &= \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}, & \text{(iii)} \quad \frac{\pi^2}{8} &= \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots \\
\text{(iv)} \quad & \text{By using Parseval's identity prove that } \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^4}{90}
\end{aligned}$$

Solution: Here $f(x)$ is an even function because $f(-x) = (-x)^2 = x^2 = f(x)$ [$\therefore b_n = 0$]

Hence, the Fourier expansion for $f(x)$ is $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx)$ (1)

$$a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) \, dx, \quad a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx \, dx \quad \text{and} \quad b_n = 0$$

$$\therefore a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) \, dx = \frac{2}{\pi} \int_0^{\pi} x^2 \, dx = \frac{2}{\pi} \left[\frac{x^3}{3} \right]_0^{\pi} = \frac{2\pi^2}{3}$$

$$\begin{aligned}
a_n &= \frac{2}{\pi} \int_0^\pi f(x) \cos nx \, dx \\
&= \frac{2}{\pi} \left[x^2 \left(\frac{\sin nx}{n} \right) - (2x) \left(-\frac{\cos nx}{n^2} \right) + 2 \left(-\frac{\sin nx}{n^3} \right) \right]_0^\pi \\
&= \frac{2}{\pi} \left[\left\{ 0 - (2\pi) \left\{ -\frac{\cos n\pi}{n^2} \right\} + 0 \right\} - \{0\} \right] \\
&= \frac{4}{n^2} \cos n\pi = \frac{4}{n^2} (-1)^n
\end{aligned}$$

Putting these values in (1) $x^2 = \frac{\pi^2}{3} + 4 \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \cos nx$

(i) For first deduction put $x = \pi$

$$\begin{aligned}
\therefore \pi^2 &= \frac{\pi^2}{3} + 4 \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \cos n\pi \\
\pi - \frac{\pi^2}{3} &= 4 \sum_{n=1}^{\infty} \frac{(-1)^n (-1)^n}{n^2} = 4 \sum_{n=1}^{\infty} \frac{(-1)^{2n}}{n^2} \\
\therefore \frac{2\pi^2}{3} \cdot \frac{1}{4} &= \sum_{n=1}^{\infty} \frac{(-1)^{2n}}{n^2} \quad \therefore \frac{\pi^2}{6} = \sum_{n=1}^{\infty} \frac{1}{n^2}
\end{aligned}$$

(ii) For second deduction put $x = 0$

$$\begin{aligned}
\therefore 0 &= \frac{\pi^2}{3} + 4 \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \cdot 1 \\
\therefore -\frac{\pi^2}{3} &= 4 \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \quad \therefore \frac{\pi^2}{12} = (-1) \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \\
\therefore \frac{\pi^2}{12} &= \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}
\end{aligned}$$

(iii) For third deduction, add the above two results

$$\begin{aligned}
\frac{\pi^2}{6} &= \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \frac{1}{5^2} + \dots \\
\frac{\pi^2}{12} &= \frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \frac{1}{5^2} - \dots \\
\therefore \frac{\pi^2}{6} + \frac{\pi^2}{12} &= 2 \left(\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots \right) \\
\therefore \frac{\pi^2}{8} &= \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots
\end{aligned}$$

(iv) By Parseval's identity $\frac{1}{\pi} \int_{-\pi}^{\pi} [f(x)]^2 \, dx = \frac{a_0^2}{2} + \sum (a_n^2 + b_n^2)$

Here, $f(x) = x^2$, $a_0 = \frac{2\pi^2}{3}$, $a_n = \frac{4(-1)^n}{n^2}$, $b_n = 0$

$$\begin{aligned}
\therefore \frac{1}{\pi} \int_{-\pi}^{\pi} x^4 \, dx &= \frac{1}{\pi} \left[\frac{x^5}{5} \right]_{-\pi}^{\pi} = \frac{2\pi^4}{5} \\
\therefore \frac{2\pi^4}{5} &= \frac{2\pi^4}{9} + 16 \left[\frac{1}{1^4} + \frac{1}{2^4} + \dots \right] \\
\therefore \pi^4 \left(\frac{1}{5} - \frac{1}{9} \right) &= 8 \left[\frac{1}{1^4} + \frac{1}{2^4} + \dots \right] \\
\frac{\pi^4}{90} &= \frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \dots
\end{aligned}$$

5. Obtain Fourier series for $f(x)$ in the range $(-\pi, \pi)$, where $f(x) = |x|$

Solution: The graph is symmetrical about y-axis, therefore, $f(x)$ is even

Hence, the Fourier expansion for $f(x)$ is $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx)$

$$a_0 = \frac{2}{\pi} \int_0^\pi f(x) \, dx, \quad a_n = \frac{2}{\pi} \int_0^\pi f(x) \cos nx \, dx \quad \text{and} \quad b_n = 0$$

$$\therefore a_0 = \frac{2}{\pi} \int_0^\pi f(x) \, dx = \frac{2}{\pi} \int_0^\pi x \, dx = \frac{2}{\pi} \left[\frac{x^2}{2} \right]_0^\pi = \pi$$

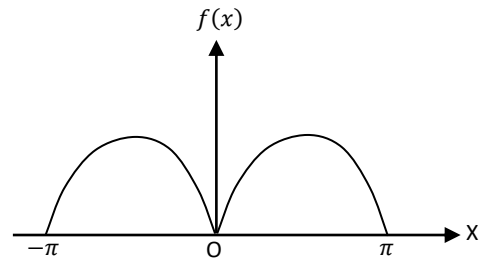
$$a_n = \frac{2}{\pi} \int_0^\pi x \cdot \cos nx \, dx$$

$$\begin{aligned}
&= \frac{2}{\pi} \left[x \cdot \left(\frac{\sin nx}{n} \right) - (1) \cdot \left(-\frac{\cos nx}{n^2} \right) \right]_0^\pi \\
&= \frac{2}{\pi} \left[\frac{\cos n\pi}{n^2} - \frac{1}{n^2} \right] \\
&= \frac{2}{\pi n^2} \cdot [(-1)^n - 1] \\
\therefore a_n &= \frac{-4}{\pi n^2} \quad \text{when } n \text{ is odd} \\
&= 0 \quad \text{when } n \text{ is even} \\
\text{Hence } a_n &= \frac{-4}{\pi(2n-1)^2} \quad \{\text{When } n \text{ is odd, it can be replaced by } (2n-1)\} \\
\text{Thus, the required Fourier Series is } f(x) &= \frac{\pi}{2} - \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} \cos(2n-1)x
\end{aligned}$$

6. Obtain a Fourier series for $f(x) = |\sin x|$ in the range $-\pi < x < \pi$

Solution: The function $f(x)$ is defined as follows:

$$\begin{aligned}
f(x) &= \begin{cases} -\sin x & \text{for } -\pi < x < 0 \\ \sin x & \text{for } 0 < x < \pi \end{cases} \\
f(-x) &= \begin{cases} -\sin(-x) & \text{for } -\pi < -x < 0 \\ \sin(-x) & \text{for } 0 < -x < \pi \end{cases} \\
&= \begin{cases} \sin x & \text{for } \pi > x > 0 \\ -\sin x & \text{for } 0 > x > -\pi \end{cases} \\
&= \begin{cases} -\sin x & \text{for } -\pi < x < 0 \\ \sin x & \text{for } 0 < x < \pi \end{cases} = f(x)
\end{aligned}$$



$\therefore f(x)$ is an even function

$$\therefore f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx)$$

$$a_0 = \frac{2}{\pi} \int_0^\pi \sin x \, dx = \frac{2}{\pi} [-\cos x]_0^\pi = \frac{-2}{\pi} [\cos \pi - \cos 0] = \frac{-2}{\pi} [-1 - 1] = \frac{4}{\pi}$$

$$\begin{aligned}
a_n &= \frac{2}{\pi} \int_0^\pi \sin x \cdot \cos nx \, dx = \int_0^\pi [\sin(1+n)x + \sin(1-n)x] \, dx \\
&= \frac{1}{\pi} \left[\frac{-\cos(1+n)x}{(1+n)} - \frac{\cos(1-n)x}{(1-n)} \right]_0^\pi \\
&= -\frac{1}{\pi} \left\{ \left[\frac{\cos(n+1)\pi}{(n+1)} - \frac{\cos(n-1)\pi}{(n-1)} \right] - \left[\frac{1}{(n+1)} - \frac{1}{(n-1)} \right] \right\} \\
&= -\frac{1}{\pi} \left\{ \left[\frac{-(-1)^n}{(n+1)} - \frac{-(-1)^n}{(n-1)} \right] - \left[\frac{1}{(n+1)} - \frac{1}{(n-1)} \right] \right\} \quad \{\text{Since } \cos(n+1)\pi = \cos(n-1)\pi = -(-1)^n\} \\
&= -\frac{1}{\pi} \left\{ \left[\frac{-(-1)^n(n-1-n-1)}{n^2-1} \right] - \left[\frac{n-1-n-1}{n^2-1} \right] \right\} \\
&= -\frac{1}{\pi} \left[\frac{+2}{n^2-1} \right] [(-1)^n + 1] \\
&= -\frac{4}{\pi(n^2-1)} \quad \text{when } n \text{ is even} \\
&= 0 \quad \text{when } n \text{ is odd} \\
\therefore a_n &= \frac{-4}{\pi(4n^2-1)} \quad \{\text{for } n \text{ even, } n \text{ is replaced by } 2n\}
\end{aligned}$$

$$\text{Thus, the required Fourier series is } f(x) = \frac{2}{\pi} - \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{\cos 2n\pi}{4n^2-1}$$

$$\text{or it can be written in the expanded form as: } f(x) = \frac{2}{\pi} - \frac{4}{\pi} \left[\frac{\cos 2\pi}{3} + \frac{\cos 4\pi}{15} + \frac{\cos 6\pi}{35} + \dots \right]$$

7. Obtain Fourier series for $f(x) = x \cos x$ in $(-\pi, \pi)$

Solution: Since, $f(-x) = (-x) \cos(-x) = -x \cos x = -f(x)$, $f(x)$ is an odd function [$\therefore a_n = 0$]

$\therefore$ Let $f(x) = x \cos x = \sum_{n=1}^{\infty} b_n \sin nx$

$$\begin{aligned}
 b_n &= \frac{2}{\pi} \int_0^{\pi} f(x) \sin nx \, dx = \frac{2}{\pi} \int_0^{\pi} x \cos x \sin nx \, dx \quad \dots\dots\dots (1) \\
 &= \frac{1}{\pi} \int_0^{\pi} x \cdot 2 \sin nx \cos x \, dx \\
 &= \frac{1}{\pi} \int_0^{\pi} x [\sin(n+1)x + \sin(n-1)x] \, dx \\
 &= \frac{1}{\pi} \int_0^{\pi} x \sin(n+1)x \, dx + \frac{1}{\pi} \int_0^{\pi} x \sin(n-1)x \, dx \\
 &= \frac{1}{\pi} \left\{ \left[x \cdot \frac{-\cos(n+1)x}{n+1} - (1) \cdot \frac{-\sin(n+1)x}{(n+1)^2} \right]_0^{\pi} + \left[x \cdot \frac{-\cos(n-1)x}{n-1} - (1) \cdot \frac{-\sin(n-1)x}{(n-1)^2} \right]_0^{\pi} \right\} \\
 &= \frac{1}{\pi} \left[\frac{\pi}{n+1} (-\cos(n+1)\pi) + \frac{\pi}{n-1} (-\cos(n-1)\pi) \right] \\
 &= \frac{(-1)^n}{n+1} + \frac{(-1)^n}{n-1} \quad [\because \cos(n\pi \pm \pi) = -\cos n\pi] \\
 &= (-1)^n \left[\frac{1}{n+1} + \frac{1}{n-1} \right] = (-1)^n \cdot \frac{2n}{n^2-1} \text{ if } n \neq 1
 \end{aligned}$$

For $n = 1$, we put $n = 1$ in (1)

$$\begin{aligned}
 \therefore b_1 &= \frac{2}{\pi} \int_0^{\pi} x \cos x \sin x \, dx = \frac{1}{\pi} \int_0^{\pi} x \sin 2x \, dx \\
 &= \frac{1}{\pi} \left[x \cdot \frac{-\cos 2x}{2} - (1) \cdot \frac{-\sin 2x}{4} \right]_0^{\pi} \\
 &= \frac{1}{\pi} \left(-\frac{\pi}{2} \right) = -\frac{1}{2}
 \end{aligned}$$

Hence, $x \cos x = -\frac{1}{2} \sin x + 2 \sum_{n=2}^{\infty} (-1)^n \cdot \frac{n}{n^2-1} \sin nx$

8. Obtain Fourier series for $f(x)$ given by $f(x) = \begin{cases} 1 + \frac{2x}{\pi}, & -\pi < x < 0 \\ 1 - \frac{2x}{\pi}, & 0 < x < \pi \end{cases}$

Hence deduce that $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots = \frac{\pi^2}{8}$

Solution: Here $f(x)$ is an even function because

$$f(-x) = \begin{cases} 1 - \frac{2x}{\pi}, & -\pi < -x < 0 \\ 1 + \frac{2x}{\pi}, & 0 < -x < \pi \end{cases} = \begin{cases} 1 - \frac{2x}{\pi}, & \pi \geq x \geq 0 \\ 1 + \frac{2x}{\pi}, & 0 \geq x \geq -\pi \end{cases} = f(x)$$

$$\therefore b_n = 0$$

$$\text{Let } f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx$$

$$\therefore a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) \, dx = \frac{2}{\pi} \int_0^{\pi} \left(1 - \frac{2x}{\pi} \right) \cdot dx = \frac{2}{\pi} \left[x - \frac{x^2}{\pi} \right]_0^{\pi} = \frac{2}{\pi} [\pi - \pi] = 0$$

$$\begin{aligned}
 a_n &= \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx \, dx = \frac{2}{\pi} \int_0^{\pi} \left(1 - \frac{2x}{\pi} \right) \cos nx \, dx \\
 &= \frac{2}{\pi} \left[\left(1 - \frac{2x}{\pi} \right) \left(\frac{\sin nx}{n} \right) - \left(-\frac{2}{\pi} \right) \left(-\frac{\cos nx}{n^2} \right) \right]_0^{\pi} \\
 &= \frac{2}{\pi} \left[\left\{ 0 - \frac{2 \cos n\pi}{\pi n^2} \right\} - \left\{ -\frac{2}{\pi n^2} \right\} \right] \\
 &= \frac{4}{\pi^2 n^2} [1 - \cos n\pi] \\
 &= \begin{cases} 0, & \text{if } n \text{ is even} \\ \frac{8}{\pi^2 n^2}, & \text{if } n \text{ is odd} \end{cases}
 \end{aligned}$$

$$\therefore f(x) = \frac{8}{\pi^2} \left[\frac{\cos x}{1^2} + \frac{\cos 3x}{3^2} + \frac{\cos 5x}{5^2} + \dots \right]$$

Now, by putting $x = 0$ in the above result, we can get $\frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots$

9. Obtain Fourier series for $f(x) = \sin hax$, in $(-\pi, \pi)$, where a is not an integer.

Solution: Here $f(x)$ is an odd function because

$$f(-x) = \sinh(-ax) = \frac{e^{-ax} - e^{ax}}{2} = -\left(\frac{e^{ax} - e^{-ax}}{2}\right) = -\sinh ax = -f(x)$$

$$\therefore a_0 = 0 \text{ \& } a_n = 0$$

$$\text{Let } f(x) = \sum_{n=1}^{\infty} b_n \sin nx$$

$$\therefore b_n = \frac{2}{\pi} \int_0^{\pi} f(x) \sin nx \, dx = \frac{2}{\pi} \int_0^{\pi} \sinh ax \sin nx \, dx$$

To find b_n , we use the result $\cos(n - ia) = \cos nx \cos iax + \sin nx \sin iax$

$$\cos(n - ia) = \cos nx \cosh ax + i \sin nx \sinh ax \quad (\because \cos ix = \cosh x, \sin ix = i \sinh x)$$

Hence,

$$\begin{aligned} b_n &= \frac{2}{\pi} \int_0^{\pi} \sin nx \sinh ax \, dx = \frac{2}{\pi} \text{I. P.} \int_0^{\pi} \cos(n - ia)x \, dx \\ &= \frac{2}{\pi} \text{I. P.} \left[\frac{\sin(n - ia)x}{n - ia} \right]_0^{\pi} = \frac{2}{\pi} \text{I. P.} \frac{\sin(n - ia)\pi}{n - ia} \\ &= \frac{2}{\pi} \text{I. P.} (n + ia) \frac{\sin(n - ia)\pi}{n^2 - i^2 a^2} = \frac{2}{\pi} \text{I. P.} \frac{(n + ia)}{n^2 + a^2} \sin(n\pi - ia\pi) \\ &= \frac{2}{\pi} \text{I. P.} \frac{(n + ia)}{(n^2 + a^2)} [\sin n\pi \cos ia\pi - \cos n\pi \sin ia\pi] \\ &= \frac{2}{\pi} \text{I. P.} \frac{(n + ia)}{(n^2 + a^2)} [-i \cos n\pi \sinh a\pi] \\ &= \frac{2}{\pi} \text{I. P.} \frac{(n + ia)}{n^2 + a^2} (-i)(-1)^n \sinh a\pi \\ &= \frac{2n}{(n^2 + a^2)\pi} (-1)^{n+1} \sinh a\pi \end{aligned}$$

Alternatively,

$$\begin{aligned} b_n &= \frac{2}{\pi} \int_0^{\pi} \frac{e^{ax} - e^{-ax}}{2} \cdot \sin nx \, dx \\ &= \frac{1}{\pi} \left[\frac{e^{ax}}{a^2 + n^2} (a \sin x - n \cos nx) - \frac{e^{-ax}}{a^2 + n^2} (-a \sin nx - n \cos nx) \right]_0^{\pi} \\ &= \frac{2n}{(n^2 + a^2)\pi} \cdot (-1)^{n+1} \sinh a\pi \end{aligned}$$

$$\therefore f(x) = \sum_{n=1}^{\infty} b_n \sin nx = \frac{2}{\pi} \sinh a\pi \sum_{n=1}^{\infty} \frac{(-1)^{n+1} \cdot n}{n^2 + a^2} \sin nx$$

10. Find the Fourier series for $f(x) = x + x^2$ in $-\pi \leq x \leq \pi$. Hence deduce that :

$$(i) \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots = \frac{\pi^2}{6}. \quad (ii) \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} \dots = \frac{\pi^2}{8}.$$

Solution: We first note that $f(x) = x + x^2$ is the sum of the odd function $f_1(x) = x$ and even function $f_2(x) = x^2$

Hence Fourier expansion of $f(x)$ is the sum of Fourier expansion of $f_1(x)$ and $f_2(x)$

Now, since $f_1(x) = x$ is an odd function $[\therefore a_0 = 0 \text{ \& } a_n = 0]$

$$\text{Let } f_1(x) = x = \sum_{n=1}^{\infty} b_n \sin nx$$

$$\begin{aligned}
b_n &= \frac{2}{\pi} \int_0^\pi f(x) \sin nx \, dx = \frac{2}{\pi} \int_0^\pi x \sin nx \, dx \\
&= \frac{2}{\pi} \left[x \left(-\frac{\cos nx}{n} \right) - (1) \left(-\frac{\sin nx}{n^2} \right) \right]_0^\pi \\
&= \frac{2}{\pi} \left[\left\{ \pi(-1) \frac{(-1)^n}{n} - 0 \right\} - \{0\} \right] = \frac{2(-1)^{n+1}}{n} \\
\therefore f_1(x) = x &= 2 \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \sin nx \quad \dots\dots\dots (1)
\end{aligned}$$

Now, since $f_2(x) = x^2$ is an even function $[\therefore b_n = 0]$

Let $f_2(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx$

$$a_0 = \frac{2}{\pi} \int_0^\pi f_2(x) \, dx = \frac{2}{\pi} \int_0^\pi x^2 \, dx = \frac{2}{\pi} \left[\frac{x^3}{3} \right]_0^\pi = \frac{2\pi^2}{3}$$

$$\begin{aligned}
a_n &= \frac{2}{\pi} \int_0^\pi f(x) \cos nx \, dx = \frac{2}{\pi} \int_0^\pi x^2 \cdot \cos nx \, dx \\
&= \frac{2}{\pi} \left[x^2 \left(\frac{\sin nx}{n} \right) - (2x) \left(-\frac{\cos nx}{n^2} \right) + (2) \left(-\frac{\sin nx}{n^3} \right) \right]_0^\pi \\
&= \frac{2}{\pi} \left[\left\{ 0 - 2\pi \left(-\frac{\cos n\pi}{n^2} \right) - 0 \right\} - \{0\} \right] = 4 \frac{(-1)^n}{n^2} \\
\therefore f_2(x) = x^2 &= \frac{\pi^2}{3} + 4 \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \cos nx \quad \dots\dots\dots (2)
\end{aligned}$$

$$\text{From (1) and (2)} \quad x + x^2 = \frac{\pi^2}{3} + 4 \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \cos nx + 2 \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \sin nx$$

For deduction, it is enough to consider the series of x^2 . We have obtained in (2)

$$x^2 = \frac{\pi^2}{3} + 4 \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \cos nx \quad \dots\dots\dots (3)$$

(i) Putting $x = \pi$ in (3)

$$\begin{aligned}
\pi^2 &= \frac{\pi^2}{3} + 4 \left[-\frac{1}{1^2} \cos \pi + \frac{1}{2^2} \cos 2\pi - \frac{1}{3^2} \cos 3\pi + \dots \right] \\
\frac{2\pi^2}{3} &= 4 \left[\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots \right] \\
\frac{\pi^2}{6} &= \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots \quad \dots\dots\dots (4)
\end{aligned}$$

Putting $x = 0$ in (3)

$$\begin{aligned}
0 &= \frac{\pi^2}{3} + 4 \left[-\frac{1}{1^2} + \frac{1}{2^2} - \frac{1}{3^2} + \dots \right] \\
\frac{\pi^2}{12} &= \frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} \dots \quad \dots\dots\dots (5)
\end{aligned}$$

(ii) Adding (4) and (5)

$$\begin{aligned}
\frac{\pi^2}{6} + \frac{\pi^2}{12} &= 2 \left(\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots \right) \\
\frac{\pi^2}{8} &= \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots
\end{aligned}$$

$$11. \quad \text{If } f(x) = \begin{cases} x + \frac{\pi}{2}, & -\pi \leq x \leq 0 \\ \frac{\pi}{2} - x, & 0 \leq x \leq \pi \end{cases} \quad \text{Prove that : } f(x) = \frac{4}{\pi} \left(\cos x + \frac{1}{3^2} \cos 3x + \frac{1}{5^2} \cos 5x + \dots \right)$$

Hence deduce that : $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots = \frac{\pi^2}{8}$ Also deduce that $\frac{\pi^4}{96} = \frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \dots$

Solution: Clearly $f(x)$ is an even function

$$f(-x) = \begin{cases} -x + \frac{\pi}{2}, & -\pi < -x < 0 \\ \frac{\pi}{2} + x, & 0 < -x < \pi \end{cases} = \begin{cases} \frac{\pi}{2} - x, & \pi > x > 0 \\ \frac{\pi}{2} + x, & 0 < x < -\pi \end{cases} = f(x)$$

$$\therefore b_n = 0$$

$$\therefore f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx$$

$$a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) dx = \frac{2}{\pi} \int_0^{\pi} \left(\frac{\pi}{2} - x\right) dx = \frac{2}{\pi} \left[\frac{\pi}{2}x - \frac{x^2}{2}\right]_0^{\pi} = 0 \quad \dots\dots\dots (1)$$

$$\begin{aligned} a_n &= \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx dx = \frac{2}{\pi} \int_0^{\pi} \left(\frac{\pi}{2} - x\right) \cos nx dx \\ &= \frac{2}{\pi} \left[\left(\frac{\pi}{2} - x\right) \left(\frac{\sin nx}{n}\right) - (-1) \cdot \left(-\frac{\cos nx}{n^2}\right) \right]_0^{\pi} \\ &= \frac{2}{\pi} \left[-(-1)^n \cdot \frac{1}{n^2} + \frac{1}{n^2} \right] \\ &= \frac{2}{\pi} \left[\frac{1}{n^2} - \frac{(-1)^n}{n^2} \right] = \frac{2}{\pi n^2} \left[\frac{1 - (-1)^n}{n^2} \right] = \begin{cases} \frac{2}{\pi n^2} & n \text{ is odd} \\ 0 & n \text{ is even} \end{cases} \end{aligned}$$

$$\therefore f(x) = \frac{4}{\pi} \left[\frac{1}{1^2} \cos x + \frac{1}{3^2} \cos 3x + \frac{1}{5^2} \cos 5x + \dots \right] \quad \dots\dots\dots (2)$$

(i) Now $f(x)$ is discontinuous at $x = 0$

But at a point of discontinuity $x = c$

$$f(x) = \frac{1}{2} \left[\lim_{x \rightarrow c^-} f(x) + \lim_{x \rightarrow c^+} f(x) \right] \quad \therefore f(0) = \frac{1}{2} \left[\frac{\pi}{2} + \frac{\pi}{2} \right] = \frac{\pi}{2}$$

$\therefore$ Putting $x = 0$,

$$\frac{\pi}{2} = \frac{4}{\pi} \left[\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots \right]$$

$$\therefore \frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots$$

(ii) We now use Parseval's identity in $(-\pi, \pi)$

$$\frac{1}{\pi} \int_{-\pi}^{\pi} [f(x)]^2 dx = \frac{a_0^2}{2} + \sum (a_n^2 + b_n^2) \quad \dots\dots\dots (3)$$

$$\begin{aligned} \therefore \frac{1}{\pi} \int_{-\pi}^{\pi} [f(x)]^2 dx &= \frac{1}{\pi} \left[\int_{-\pi}^0 \left(x + \frac{\pi}{2}\right)^2 dx + \int_0^{\pi} \left(\frac{\pi}{2} - x\right)^2 dx \right] \\ &= \frac{1}{\pi} \left[\int_{-\pi}^0 \left(x^2 + \pi x + \frac{\pi^2}{4}\right) dx + \int_0^{\pi} \left(\frac{\pi^2}{4} - \pi x + x^2\right) dx \right] \\ &= \frac{1}{\pi} \left[\left\{ \frac{x^3}{3} + \frac{\pi x^2}{2} + \frac{\pi^2}{4} x \right\}_{-\pi}^0 + \left\{ \frac{\pi^2}{4} x - \frac{\pi x^2}{2} + \frac{x^3}{3} \right\}_0^{\pi} \right] \\ &= \frac{1}{\pi} \left[0 - \left\{ -\frac{\pi^3}{3} + \frac{\pi^3}{2} - \frac{\pi^3}{4} \right\} + \left\{ \frac{\pi^3}{4} - \frac{\pi^3}{2} + \frac{\pi^3}{3} \right\} - 0 \right] \\ &= \frac{2\pi^3}{\pi} \left[\frac{1}{4} - \frac{1}{2} + \frac{1}{3} \right] = \frac{\pi^2}{6} \end{aligned}$$

Hence, from (1), (2) and (3),

$$\frac{\pi^2}{6} = \frac{16}{\pi^2} \left[\frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \dots \right]$$

$$\frac{\pi^4}{96} = \frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \dots$$

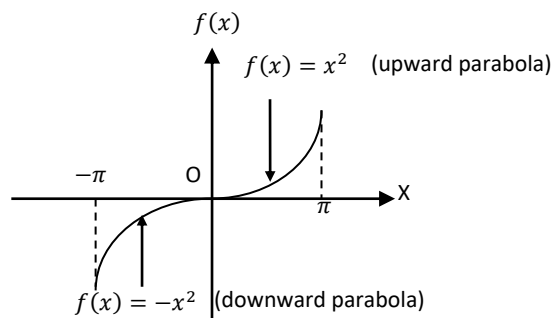
12. Obtain Fourier series for $f(x)$ given by $f(x) = \begin{cases} -x^2, & -\pi < x < 0 \\ x^2, & 0 < x < \pi \end{cases}$

Solution: $f(x) = \begin{cases} -x^2, & -\pi < x < 0 \\ x^2, & 0 < x < \pi \end{cases}$

$$f(-x) = \begin{cases} -(-x)^2, & -\pi < (-x) < 0 \\ (-x)^2, & 0 < (-x) < \pi \end{cases}$$

$$f(-x) = \begin{cases} -x^2, & \pi > x > 0 \\ x^2, & 0 > x > -\pi \end{cases}$$

$$f(-x) = \begin{cases} x^2, & -\pi < x < 0 \\ -x^2, & 0 < x < \pi \end{cases} = -\begin{cases} -x^2, & -\pi < x < 0 \\ x^2, & 0 < x < \pi \end{cases} = -f(x)$$



$\therefore$ the given function is odd

This is also clear from the following graph, which is symmetrical about the origin

Hence in the Fourier expansion $a_0 = 0$, $a_n = 0$, $b_n = \frac{2}{\pi} \int_0^\pi f(x) \cdot \sin nx \, dx$

and the Fourier series is $f(x) = \sum_{n=1}^{\infty} (b_n \sin nx)$

$$\therefore b_n = \frac{2}{\pi} \int_0^\pi x^2 \cdot \sin nx \, dx$$

$$= \frac{2}{\pi} \left[x^2 \cdot \left(-\frac{\cos nx}{n} \right) - (2x) \cdot \left(-\frac{\sin nx}{n^2} \right) + (2) \left(\frac{\cos nx}{n^3} \right) \right]_0^\pi$$

$$= \frac{2}{\pi} \left[\pi^2 \cdot \left(-\frac{\cos n\pi}{n} \right) + (2) \cdot \left(\frac{\cos n\pi}{n^3} \right) - 2 \cdot \left(\frac{1}{n^3} \right) \right]$$

$$= \frac{-2\pi(-1)^n}{n} + \frac{4}{n^3\pi} \cdot [(-1)^n - 1]$$

$$\therefore b_n = \frac{2\pi}{n} - \frac{8}{n^3\pi} \quad \text{when } n \text{ is odd}$$

$$= \frac{-2\pi}{n} \quad \text{when } n \text{ is even}$$

Hence the required Fourier Series is

$$f(x) = 2 \left(\pi - \frac{4}{\pi} \right) \sin x - \pi \cdot \sin 2x + 2 \left(\frac{\pi}{3} - \frac{4}{27\pi} \right) \sin 3x - \frac{\pi}{2} \cdot \sin 4x + \dots$$